Four Tests of the Cell System

Paper 10. Jacobsthal, almost-primes between squares, shared cofactors, and collisions

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Abstract

Four tests of the same framework, each carried to the point where it stops.

Shared cofactors (Theorems 1 and 2). If odd primes $p < q$ and an integer $a \geq q$ satisfy

$$k^2 < pa < qa < (k+2)^2 \quad \text{for some odd } k,$$

then $q = p + 2$. Every twin produces such a sharing, so infinitely many twins exist if and only if sharing occurs at unbounded height.

Collisions (Theorems 3 and 4). For odd $p > 7$, the collision count E_p of the sharing set satisfies

$$E_p = 0 \iff p + 2 \text{ is prime},$$

so the criterion reads the primality of $p + 2$ rather than assuming it. For $p > 121$ the witness lies in the belt $\sqrt{p+2} < v \leq (p+11)/6$, and in a narrow slice at its top the test collapses to divisibility by 3 (Theorems 5 and 6).

Jacobsthal's function is reproduced exactly on the cycle and yields no bound (§2.2–2.4). **Almost-primes between squares** give an asymptotic pair statement with no parity cost; an explicit version fails at a measured log-lag wall (§2.5–2.7).

Each test separates its exact statements from its measured diagnostics. **None of the four closes the gap, and the paper says in each case why.**

Numbering. This paper is one part of a set that was written as a single document and is now published in parts. Each part numbers its own results from one and is self-contained: a reference of the form [Pn, Thm 1] means Theorem 1 of Paper n, and an unqualified "Theorem 1" always means this paper's own. Result numbers therefore differ from those of the earlier seven-document releases, where the whole set shared the numbering of the single document.

How to read the claims in this paper. Statements set as Theorems, Propositions and Corollaries are proved, and the proofs are given. Anything described as *measured* is a computation over a stated finite range and is labelled as such where it occurs.

Keywords: twin primes, Jacobsthal function, almost-primes, difference of squares, cell coordinates.

MSC 2020: 11N35, 11N05, 11A41.

Summary of the results in this part

§2 — Four tests, the narrowest form of the requirement, and the pattern of the whole

Result	What it says	Section
Proposition 1	*Measured, not proved.* Saturation of extremal runs in the first test, Jacobsthal's function — and §2.4 explains why it yields no bound.	§2.3
Second test	Almost-primes between squares, with the correlation of the survivor count and the order of its sum.	§2.5–2.6
The narrowest form	The requirement reduced as far as the construction can take it — and why it is still the wall.	§2.7
The first moment	Why a mean below one would suffice, and why no upper bound on it is available here.	§2.13
Theorem 1	Two odd primes sharing a cofactor inside one window must differ by 2 — no primitivity, no primality of a, and $a \geq q$ is automatic.	§2.8
Theorem 2	The equivalence: infinitely many twins iff sharing occurs at unbounded height. The witness $a = p + 6$ works for every twin.	§2.8
Theorem 3	$E_p = 0$ iff $p + 2$ is prime: the classical difference-of-squares witness always lies inside the sharing set S_p.	§2.11
Theorem 4	A witness can always be taken in the belt $\sqrt{p+2} < v \leq (p+11)/6$, for every odd $p > 121$ (below 121 there are eight exceptions).	§2.12
Theorem 5	In a slice of width below $\sqrt{N/3}$ at the top of the belt the test collapses to divisibility by 3.	§2.12
Theorem 6	Collisions enter the slice in stages $t = \frac{1}{2}(d - 3k)$ with $t \leq \lfloor 3(L - \frac{3}{2})^2/2N \rfloor$, widening Theorem 5 by $\sqrt{2}$.	§2.12
Theorem 7	Inside one list the value of $\Delta = (N - 6v)^2 + 48t$ determines the collision, so counting Δ is not a relaxation.	§2.14

1. Setting

[P1] to [P8] supply the framework under test. [P9] gives the exact form of the obstruction each test is measured against, and [P11] collects the routes closed elsewhere. Nothing else is imported.

- **[P1]** supplies the window combinatorics: the increments W_j of $\lfloor 2j^2/p \rfloor$, their uniform bound and their exact histogram.
- **[P2]** supplies the coordinates: the line $L_p(k) = p(p + 2k)$ beginning at p^2, the grid L_3, the cells $C_b = (6b - 1, 6b + 1)$.
- **[P3]** supplies the exact cycle laws: the four-state census and its refinement by inheritance depth.
- **[P4]** supplies the passage to a short window, and the measurement $T/M \approx 0.80$ at moving depth.
- **[P5]**–**[P8]** supply what this paper tests: the gap alphabet, the six exception positions, the closing budget, the clocks, and the sector inheritance laws.

Three words used throughout, as defined in [P2]. A **line** L_p is the odd multiples of p from p^2 onward. A **cell** is a pair $C_b = (6b - 1,\ 6b + 1)$; it is **open**, or **survives** a set of lines, when neither member lies on any of them. The **window** is the interval between two consecutive odd squares; in

cell coordinates it is a block of consecutive indices. A twin pair is an open cell with both members prime; inside (u^2, v^2) sieved by every line up to u the two notions coincide [P6, §2.1].

Organisation. §2 tests the framework on four problems whose answers are known independently of it: Jacobsthal's function (§§2.2–2.4), almost-primes between consecutive squares (§§2.5–2.6), a shared cofactor between two lines (§§2.8–2.10), and collisions inside the sharing set (§§2.11–2.14). §2.7 states the narrowest form of the twin requirement, and §2.15 the pattern common to all four. A statement labelled *measured* is a numerical finding with its controls, not a theorem and not an asymptotic claim.

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2. Four tests, the sharpest form of the requirement, and the pattern of the whole

Each case is a question the framework can pose in its own coordinates, whose answer is already known, or known to be hard, by other means. So each is a test of the framework, not an application. §2.1 states the pattern they share, as a summary of the tested cases, not a universal classification. §§2.2–2.4, §§2.5–2.6, §§2.8–2.10 and §§2.11–2.14 run the four cases; §2.7 gives the narrowest form of the twin requirement in this work, and §2.15 the common pattern.

2.1 The pattern seen in the test cases

periodic statement / zero prime input	decided internally in these examples	gap 2
one prime in a progression	Dirichlet supplies the missing input here	gap 4
two simultaneous primes	**not decided by the present framework**	gap 6 / twins

The table summarises the examples studied here; it is not a theorem classifying every prime problem. The framework describes the periodic divisibility pattern exactly. Simultaneous primality needs information those periodic counts do not contain.

2.2 A test case: Jacobsthal's function

Jacobsthal's function $j(n)$ [11], the maximal gap between integers coprime to n, is such a statement: no primality enters, only the pattern modulo a primorial. Computed directly from the pattern, $j(P_k) = 2, 4, 6, 10, 14, 22, 26$ for $k = 1, \ldots, 7$ (OEIS A048670 [16]) over cycles $2, 6, 30, 210, 2310, 30030, 510510$.

The literature states the bounds in the **sieving bound** y, not in $k = \pi(y)$. The translation costs a factor of log, so we give both. $P(y)$ is the product of the primes up to y, and $\log_2, \log_3$ are iterated logarithms.

	in y	in k, via $y \approx k \log k$
upper bound (Iwaniec [10], via the linear-sieve error term [9])	$j(P(y)) \ll y^2$	$\ll (k \log k)^2$
lower bound (Ford–Green–Konyagin–Maynard–Tao [8])	$j(P(y)) \gg y \log y \log_3 y / \log_2 y$	$\gg k \log^2 k \log_3 k / \log_2 k$
conjecture (Maier–Pomerance [15])	$j(P(y)) \ll y(\log y)^{2+o(1)}$	$\ll k(\log k)^{3+o(1)}$

Against the two shapes, from the tabulated values [16] for $k \le 64$: $j/(k \log^2 k) = 1.081,\ 0.868,\ 0.969,\ 0.951,\ 0.988,\ 0.996$ at $k = 5, 10, 20, 30, 40, 50$, flat at 1. But $j/(k \log k)^2 = 0.216,\ 0.087,\ 0.048,\ 0.032,\ 0.025,\ 0.020$ falls *exactly* like $1/k$, the ratio of the two shapes.

Two readings must be kept apart. (i) The data sit at $k \log^2 k$: at the FGKMT lower bound up to the factor $\log_3 k / \log_2 k$, and about a factor $\log k$ *below* the Maier–Pomerance conjecture. So the numbers do not confirm that conjecture; they only fail to contradict it. (ii) The proved upper bound is off the data by a factor of order k, not $\log^2 k$, as the clean $1/k$ decay of the second row records.

2.3 Proposition 1 (saturation of extremal runs)

Applied to the *structure*, not the *size*, of an extremal configuration, the framework shows a clean measured pattern.

> **Proposition 1 (measured).** In the maximal closed run of length L for the first k primes, every line p covers exactly $\lceil L/p \rceil$ positions, with the single exception of the largest line, which covers exactly one.

primes	L	measured coverage	ceilings $\lceil L/p \rceil$
$\{2,3\}$	3	2, 1	2, 1
$\{2,3,5\}$	5	3, 2, 1	3, 2, 1
$\{2,3,5,7\}$	9	5, 3, 2, **1**	5, 3, 2, **2**
$\{2,3,5,7,11\}$	13	7, 5, 3, 2, **1**	7, 5, 3, 2, **2**
$\{2,3,5,7,11,13\}$	21	11, 7, 5, 3, 2, **1**	11, 7, 5, 3, 2, **2**

In every displayed extremal run, no line wastes a strike relative to the ceiling pattern. The *critical* positions (covered by exactly one line) number 3/3, 4/5, 7/9, 10/13, 16/21: their count rises while their share falls slowly, from 1 to 0.76.

2.4 And why it yields no bound

Saturation feeds the counting condition $\sum_p \lceil L/p \rceil \ge L$. Since $\sum_p 1/p$ diverges, it holds for L into the tens of thousands from $k = 3$ onward:

k	3	5	10
true $j(P_k) - 1$	5	13	45
counting bound	$> 2 \times 10^4$	$> 2 \times 10^4$	$> 2 \times 10^4$

Already at small k the bound exceeds the truth by factors of 400 to 4000. **This is the failure recorded in [P11, App. B.2.2] and [P11, App. B.2.1]: the framework's local statements are exact, and summing over lines destroys them.**

2.5 A second test case: almost-primes between squares

§2.1 suggests a second family of test problems, structurally closer than Jacobsthal: *every interval* $(n^2, (n+1)^2)$ *contains an integer with* $\Omega(m) \le k$. They fit here because Ω **is precisely inheritance depth**: a point of multiplicative depth k is where k layers of inheritance met.

The literature. Brun [4] obtained $k = 11$ for large n; Chen [6] later obtained $k = 2$ for sufficiently large n. Explicit statements for *every* n are much harder. Dudek and Johnston [7] reached $k = 4$ with Kuhn's weights [14]. Campbell [5] reached $k = 3$, combining a finite verification for $n^2 \le 10^{31}$ with Richert's logarithmic weights [18] and the explicit linear sieve of Bordignon, Johnston and Starichkova [3]. Between consecutive **cubes**, Johnston, Sorenson, Thomas and Webster [12] obtain $k = 2$ for all n. Legendre's conjecture, $k = 1$, is open even under the Riemann hypothesis [17].

The framework reproduces the squares-versus-cubes contrast at the level of exponents. A survivor of sieving to depth x^θ near x has $\Omega < 1/\theta$, that is $\Omega \le \lceil 1/\theta \rceil - 1$. A window of length x^α enters through the sieve variable $s = \alpha/\theta$:

window	$\theta = 1/4$	$\theta = 1/3$	$\theta = 1/2$
squares, $\alpha = 1/2$	2.000	1.500	**1.000**
cubes, $\alpha = 2/3$	2.667	2.000	**1.333**

At the depth needed for $\Omega \le 2$, the cube window's sieve variable is larger by exactly the exponent gap $2/3 - 1/2 = 1/6$, i.e. by 33%. **This gap is one structural reason the cube problem has more sieve room than the square problem; it is not, by itself, a proof of the published distinction.** (Measured, the distinction is about provability, not truth: the minimum of Ω over the window is 1 for both squares and cubes at $x = 10^6, 10^8, 10^{10}$.)

A statement of the same shape that the framework does reach, because it asks for no primality. The obstruction of [P9, §2] is about primes, not pairs of almost-primes. Asking for *few prime factors* instead of *prime* in both members removes the parity problem entirely: a lower bound for $\Omega \le k$ with k large enough is a dimension-two sieve problem above its threshold. For every sufficiently large m this gives a pair $(n, n+2)$ inside $(m^2, (m+2)^2)$ with $\Omega \le 8$ on each member, and a count $\gg m/\log^2 m$. (That is the unweighted bound. With Richert-type weights the set does better: [P12, Thm 1] gives $\Omega \le 4$ on each member in every large window $(m^2, (m+1)^2)$, and [P12, Thm 2] a total of at most 7.)

Proved versus measured. The bound above is what a dimension-two sieve delivers with losses carried honestly; **it is not what the counts are.** Measured over the window $(m^2, (m+2)^2)$:

m	window length	both $\Omega \le 2$	both $\Omega \le 3$	both $\Omega \le 8$	$m/\log^2 m$
101	408	66	138	202	4.7
1,001	4,008	375	1,114	1,998	21.0
5,001	20,008	1,388	4,703	9,982	68.9
10,001	40,008	2,513	8,767	19,954	117.9
20,001	80,008	4,454	16,512	39,880	203.9

(A pair is counted when both n and $n+2$ lie strictly inside the window.)

The $\Omega \le 8$ count is close to *every* odd integer of the window: about $2m$, linear in m, and near 170 times the proved lower bound at $m = 10^4$. Minima over a range of m: for $m \in [10^4, 10^4 + 200)$ the least counts are 2,415 at $\Omega \le 2$, 8,700 at $\Omega \le 3$ and 19,952 at $\Omega \le 8$; for $m \in [10^3, 10^3 + 200)$ they are 353, 1,087 and 1,998.

So the gap is between a theorem and a count, not between a count and the truth. Even a pair of semiprimes at distance 2 in every such window, a far stronger statement, appears to hold with thousands of witnesses and a count growing like $m/\log m$. **We do not prove it; it is a measurement.** What the framework cannot do is push k down to 1: that is the twin statement and the wall of [P9, §2].

Where the framework stops, and where it does not. The sieve variable is $s = 1.500$ for $k = 3$ and $s = 1.000$ for $k = 2$, both far below the linear sieve's lower-bound threshold $s > 2$. Published proofs bridge the gap with **weights**. Paper 3 shows the framework carries those weights exactly on the cycle, Richert's included once the generating function is refined by the size of the line [P3, Cor 4]. Paper 4 finds that weights of that soft shape transfer very accurately on the tested windows [P4, Prop 1]. **So the experiments do not point to the algebraic form of the weight as the bottleneck; the rigorous lower-bound argument remains external.**

How much is lost in that transfer, measured. We compare the cycle law of [P3, Thm 2] with a direct census over consecutive square windows $[M^2, (M+2)^2]$ near X, sieved at depth $z = X^{1/u}$, in three layers.

Layer 1 — the mean is exact. The mean inheritance depth is $\sum_{q \le z} 2/q$, and each line strikes exactly $2/q$ of the cells of any interval, up to a bounded edge term. The measured window/cycle ratio is 1.0000 at $u = 2, 3, 4$.

Layer 2 — the state totals move by a known constant. At $u = 2$ the window's NN total, divided by its cycle value, converges to the two-dimensional Buchstab factor $e^{2\gamma}/4 = 0.793055$ of [P11, App. A]:

X	10^6	10^7	10^8	10^9	10^{10}
NN window / NN cycle	1.0113	0.9814	0.8653	0.8071	**0.7953**
distance to $e^{2\gamma}/4$	+0.218	+0.188	+0.072	+0.014	**+0.002**

Three decimal places at $X = 10^{10}$: this layer is a constant the framework already computes, not a free parameter.

Layer 3 — the shape of the depth distribution has a residual that does not decay. Within the state OO, the window's conditional mean depth is below its cycle value by +0.18 per cent, −0.24 per cent, −1.51 per cent, −1.
at the same five heights. The total-variation distance of the conditional law is 0.012, 0.028, 0.042, 0.046, 0.045. **Both saturate; neither is a finite-size effect.** Rescaling each state to its measured total removes almost none of it ($0.0770 \to 0.0755$ at $X = 10^8$), so the correction is not one multiplicative constant. Across u the residual is signed: negative on $(2, 2.7)$ with a minimum near $u = 2.2$, positive on $(2.7, 3.3)$, and indistinguishable from zero for $u \geq 3.5$. The shape is consistent with a Buchstab-type delay structure in the depth variable, but we do not identify it.

The quantity the proofs use transfers far better than the law it is computed from. Take Richert's weight, with sifting range and weight range both scaled with X ($z = X^{1/4}$, $y = X^{1/2}$), and compare the cycle prediction with the window census:

X	z	y	mean w, window / cycle	$\sum \max(0, w)$, window / cycle
10^8	100	10^4	0.99860	0.99610
10^9	177	3.2×10^4	0.99886	0.99628
10^{10}	316	10^5	0.99892	**0.99638**

The truncated sum, which a weighted sieve argument evaluates, transfers with a **relative error of 0.36%, stable across three decades**. The linear part is off by 0.11%; truncation triples this, as expected of a nonlinear functional. That is five times smaller than the 1.8% distortion of the underlying depth law. It is the phenomenon of [P4, Prop 1]: soft weights transfer, sharp indicators do not; OO is sharp, w is soft.

What this does and does not settle. A weighted-sieve argument needs the *sign* of the truncated sum, not four figures, and a third of a percent cannot change a sign. **So the computational side of the weight is not the obstruction, by a wide margin; this is now measured, not an impression.** What is missing is a *proof*: a rigorous upper bound for $\left|\sum_{\text{window}} \max(0, w) - \sum_{\text{cycle}} \max(0, w)\right|$, valid for all X, not just observed over three decades. That is the distribution of a weighted Ω in a short interval: the external ingredient of [P11, §2.1], in its most concrete form in this work.

A sharper localisation of what is missing, inside the framework. The bound above ($s = 1.500$ for $k = 3$) sieves deep enough to make survivors P_3 directly. Instead, sieve *less* deeply and let the window size supply the almost-prime bound. With $Y = \sqrt{n+1}$, sieve $(n^2, (n+1)^2)$ only to depth $Y/2$. Every prime factor of a survivor exceeds $Y/2$, the survivor is below Y^4, and $(Y/2)^5 > Y^4$ once $Y > 32$; hence

$$\Omega(x) \leq 4 \quad \text{for every survivor, once } Y > 32,$$

so the *only* obstruction to $\Omega \le 3$ in the window is a survivor with exactly four prime factors, which is tightly constrained. Write $x = pqrs$ with $p \le q \le r \le s$:

- **the fourth factor is forced.** Given p, q, r, the admissible s lies in a window of width $(2n+1)/pqr < 2Y^2/(Y/2)^3 = 16/Y$, below 1 once $Y > 16$, so there is at most one candidate;
- **the first two are confined to about four positions.** Given r, s, the product pq lies in a window of width $(2n+1)/rs < 2Y^2/(Y/2)^2 = 8$;
- **the factorisation always straddles n:** $pq \le n < rs$. This is forced: $pq \le rs$ gives $(pq)^2 \le x < (n+1)^2$ and $(rs)^2 \ge x > n^2$. (All 78 cases found at $Y = 29, 53, 101, 199, 293, 401$ agree.)

Y	29	53	101	199	293	401
survivors	321	922	2,840	9,535	19,093	33,379
with $\Omega = 4$	1	3	6	15	25	28
fraction with $\Omega \le 3$	0.99688	0.99675	0.99789	0.99843	0.99869	**0.99916**
bound $16/Y$ (widest s-window)	0.552	0.302	0.158	0.080	0.055	0.040
widest s-window measured	0.245	0.202	0.083	0.040	0.036	0.023

This makes the localisation informative rather than encouraging. So rigid an exceptional set invites an injection sending each $\Omega = 4$ survivor to a nearby $\Omega \le 3$ one. Four constructions were tried: sliding the large half, reflection in the window, replacing the least factor by a neighbouring prime, and perturbing the difference-of-squares representation. The best worked on 12 of 61 exceptional survivors, far from the required 100%. The measurement above shows why an injection was the wrong tool: **the exceptional set does not appear to be the difficulty.** At $Y = 401$ it is 0.08% of the survivors and falling. Almost all of the statement is a positive *lower bound* on the number of survivors of sieving to depth $Y/2$ in a window of length $2n+1$: a dimension-one sieve question with sieve variable

$$s = \frac{\log(2Y^2)}{\log(Y/2)} \longrightarrow 2,$$

i.e. the critical point again. **The reduction is genuine and sharp: it removes every degree of freedom but one. The one left is exactly the lower bound the framework cannot supply, at exactly the value of s where it never can.** This is §2.1's pattern at its most compressed: the framework localises the difficulty to a set of density 10^{-3} and still cannot cross it.

2.6 The correlation of the survivor count, and the order of its sum

The variance of the survivor count, and where that question already stands. Beyond the mean, one may ask for the second moment of $B(a)$, the number of cells of $(a^2, (a+2)^2)$ surviving all lines $p \le a/2$. Expanding the square brings in the correlation of the survivor indicator at cell-distance h, and **that correlation is already transported exactly by [P3, Thm 4]**. Only a change of coordinates is needed, recorded here because this section uses the resulting form.

Paper 3 indexes gap-2 pairs by x, so consecutive indices differ by 2 in the integer, and obtains the ladder $K_q(h_{\mathrm{II}}) = q-2, q-3, q-4$ according as $h_{\mathrm{II}} \equiv 0, \pm 1$, or otherwise modulo q. Cells are spaced 6 apart, so a cell-distance h is $h_{\mathrm{II}} = 3h$. Then $h_{\mathrm{II}} \equiv 0$ becomes $p \mid h$ for $p > 3$, and $h_{\mathrm{II}} \equiv \pm 1$

becomes $3h \equiv \pm 1$, i.e. $p \mid 9h^2 - 1$. Writing $\nu_p(h) = p - K_p$ for the number of residues the pair of cells forbids:

$$\nu_p(h) = \begin{cases} 2, & p \mid h, \\ 3, & p \mid 9h^2 - 1, \\ 4, & \text{otherwise}, \end{cases}$$

The three cases are mutually exclusive, since $p \mid h$ forces $3h \equiv 0$. (Directly: with $c_p \equiv 6^{-1}$, the forbidden residues for m are $\pm c_p$ and $-h \pm c_p$, which coincide in pairs under exactly those two conditions. A direct count agrees with no mismatch for all $p < 200$, $h < 300$.) **[P3, §6.3] already names the resulting product as the singular series of the Hardy–Littlewood k-tuple conjecture, "suitably normalised"; added below are the normalising constant, the exact mean, and the order of the error.** The series is

$$S_B(h; z) = \prod_{5 \le p \le z} \frac{1 - \nu_p(h)/p}{(1 - 2/p)^2} = G(z) \prod_{p \mid h} \frac{1 - 2/p}{1 - 4/p} \prod_{p \mid 9h^2 - 1} \frac{1 - 3/p}{1 - 4/p}, \qquad G(z) = \prod_{5 \le p \le z} \frac{1 - 4/p}{(1 - 2/p)^2},$$

with G convergent (log of the factor is $\asymp -4/p^2$) and $G(\infty) = 0.39688$ to five places.

This series is self-normalising, exactly. Two derivations agree. Per prime, the three cases have densities $1/p$, $2/p$ and $1 - 3/p$, so with $q = 1/p$ the expected local factor is

$$(1 - 3q) + q\frac{1 - 2q}{1 - 4q} + 2q\frac{1 - 3q}{1 - 4q} = \frac{1 - 4q + 4q^2}{1 - 4q} = \frac{(1 - 2/p)^2}{1 - 4/p},$$

the reciprocal of the matching factor of G; hence $G \prod_p E_p = 1$. Equivalently, expanding by Dirichlet convolution with $f(p) = 2/(p - 4)$ and $g(p) = 1/(p - 4)$, the condition $e \mid 9h^2 - 1$ contributes $2^{\omega(e)}$ residues and the main term is $GH \prod_p (1 + 4/(p(p - 4))) = GH \prod_p (p - 2)^2/(p(p - 4)) = H$. This is the analogue, for the pattern $h(3h - 1)(3h + 1)$, of the singular-series averages of Montgomery and Soundararajan. It identifies the object: S_B is the Hardy–Littlewood series of a prime quadruple, normalised by the square of the twin constant. Writing $\mathcal{H}_h = \{-1, 1, 6h - 1, 6h + 1\}$, the local factors at 2 and 3 are 8 and 27/16, and $\prod_{p>3}$ of the C_2 factor is $\frac{4}{3}C_2$, so that

$$\mathfrak{S}(\mathcal{H}_h) = \tfrac{27}{2}\left(\tfrac{4}{3}\right)^2 C_2^2 S_B(h) = 24 C_2^2 S_B(h).$$

The error in that average is larger than it first looks; we record the correction. An earlier draft fitted $\sum_{h \le H}(S_B(h) - 1)$ over $H \le 4 \times 10^5$, got a coefficient near 0.85 that drifted with the fitting range, and reported the error as $O(\log H)$. Extending to $H = Q_7 = 37{,}182{,}145$ (the seventh primorial from 5) shows that reading is wrong: the drift was not an unstable coefficient but the curvature of a quadratic seen over too short a range. For the smoothed sum, which fluctuates far less,

$$\Sigma_B(C) := \sum_{h < C} \Big(1 - \frac{h}{C}\Big)(S_B(h) - 1) = -0.0329(\log C)^2 - 0.244 \log C - 0.38,$$

with root-mean-square residual 0.0018 over nine points spanning $10^5 \le C \le 3.7 \times 10^7$. On the same points the linear model $a \log C + b$ has residual 0.105, and $a \log C \log\log C + \cdots$ has 0.0045. The unsmoothed sum fluctuates by $O(1)$ and cannot separate the models alone, but its local slope over the upper range is near 1.5, against the 0.85 of the short-range fit. **On this range the data favour the order $(\log C)^2$, and the earlier $O(\log H)$ is withdrawn.** The coefficient is measured, not derived; we have no closed form for it.

One consequence turns a measured constant into an explained one. In the independent-window model the dispersion is

$$\frac{\mathrm{Var}(B)}{\mathbb{E}B} = 1 - \rho_1 + 2\rho_1 \Sigma_B(C), \qquad \rho_1(z) = \prod_{5 \le p \le z} \Big(1 - \frac{2}{p}\Big) \sim \frac{12 C_2 e^{-2\gamma}}{\log^2 z} = \frac{2.497}{\log^2 z},$$

the factor 2 accounting for $\pm h$. Had Σ_B been $O(\log C)$, the product $\rho_1 \Sigma_B$ would vanish and the dispersion would tend to 1. With $\Sigma_B \asymp -c(\log C)^2$ the two logarithms cancel, and the leading term contributes $-2 \times 2.497 \times 0.0329 = -0.164$, a constant. **So, if the quadratic order is right, the sub-Poisson behaviour is a limit and not a slowly vanishing correction; the order is the measured part of this statement.**

At the measured scales, however, the subleading terms of Σ_B dominate that limit, so the model predicts a drift rather than a fixed number. Evaluating the fitted Σ_B at $z = a/2$, $C = 2a/3$:

a	10^4	10^5	10^6	10^8	$\to \infty$
ρ_1	0.0344	0.0213	0.0145	0.0079	0
model Var/$\mathbb{E}$	0.616	0.674	0.708	0.747	**0.836**
measured	0.685	0.702	—	—	—

A direct evaluation at $a = 100001$ from the exact truncated series ($C = 66{,}668$, $z = 50{,}000$, $\rho_1 = 2.13 \times 10^{-2}$, $\sum_{h \ne 0} = -13.95$) gives 0.681 against the measured 0.702, a 3% agreement with no free parameter. **The negative sign is the substance: the repulsion encoded in ν_p makes the count sub-Poisson.** The model runs a few percent low at these sizes and predicts a slow rise toward 0.836. The two measurements do rise, 0.685 at $a \sim 10^4$ to 0.702 at $a \sim 10^5$, against a predicted 0.616 to 0.674. The increments agree better than the levels, consistent with a constant offset from the Buchstab correction, which the independence model omits. We have not isolated it, and the rise remains a prediction, not a confirmed law.

Why this is a description and not a route. One might hope that a second-moment bound of the shape $\sum_{A<a\le 2A}(B(a) - \mu(a))^2 \ll \sum_{A<a\le 2A} \mu(a)$ would give $B(a) > 0$ for almost all a, far weaker than a large-deviation estimate. But that is already known, much more strongly, by other means. Bazzanella [1] proves **unconditionally** that all intervals $[n^2, (n+1)^2] \subset [3, N]$ with at most $O(N^{1/4+\varepsilon})$ exceptions contain the *expected number of primes*, not merely a survivor of the small lines. The exceptions drop to $O(f(N)\log^2 N)$ on the Riemann hypothesis, $O(N^\varepsilon)$ under the Lindelöf or Density hypotheses [2], and none under a strong form of Montgomery's conjecture. The mechanism is zero-density estimates for ζ with a fourth-power mean of ψ in short intervals. The bridge to square windows: consecutive square windows are spaced further apart than the block of exceptions forced around any one of them, so their exceptional blocks are disjoint [2, Lem. 2.2]. **So the local law above is a description of the dispersion, not the first step of a route: the**

almost-all statement it was reaching for is settled, more strongly, by analytic methods this framework does not contain.

A methodological caution that cost two false readings here. The subtracted term $W(C)e_k(\{\lambda_p\})$ is of size C, so an unclosed tail in the expectation is amplified by C. Two computations differing only in the prime bound used for G and e_k disagreed by 0.1 at $C = 1.6 \times 10^6$, more than several of the effects discussed. The disagreement was traced exactly to $4/(P \log P)$ summed over the missing primes. Any reproduction must close that tail analytically or fix the bound; the accompanying script does the former.

2.7 The narrowest form of the requirement, and why it is still the wall

The narrowest the requirement gets, and what that shows. [P6, §2.2] reduces a twin in a sector to a single open cell avoiding six named positions. Two facts sharpen that as far as the construction can. First, the lines 5 and 7 alone cut the six to three, and no further small line lowers it: a larger line can pick a phase striking none of the six, and the Chinese remainder theorem combines that freely with the phase of 5 and 7 that leaves three. Second, three open positions occur **only** when $(M + 2, M + 4)$ is itself a twin [P6, Cor. 1], so under the negation the count is at most two. And in five residue classes,

$$M \equiv 3,\ 51,\ 141,\ 153,\ 201 \pmod{210},$$

those two lines close all six, so there **every** open cell is a twin: the requirement becomes the bare $C_M \geq 1$.

That is the weakest *numerical* form of the twin requirement in this work; condition (2.3) of [P6, §2.2] is weaker but asks for no count. Three measurements say the reduction is not paid for elsewhere. Over the 713 such sectors below $M = 30{,}000$ there are 444,958 open cells and **not one fails to be a twin**, against 155 non-twin survivors in the 428 ordinary sectors below $M = 3000$ alone. The survivor count is not depressed: the density in these five classes against all others is 0.934 up to $M = 4000$, 0.997 up to 12000 and 1.000 up to 30000 — **fixing the phases of** 5 **and** 7 **changes which cells die, not how many.**

That last sentence names a standard device, not an observation of ours. Passing to a residue class modulo a primorial to remove the local bias at the small primes, at no asymptotic cost, is the W**-trick**; the modulus here is $210 = 2 \cdot 3 \cdot 5 \cdot 7$. In the analytic literature the compensating factor $\phi(W)/W$ is inserted by hand and the absence of cost is known; here the three ratios above measure it. See its use in Tao and Teräväinen [20] and in the quantitative polynomial Szemerédi work of Krause, Mousavi, Tao and Teräväinen [13]. We record the identification because this has happened repeatedly here: an object built inside the construction turns out to have a name. And C_M is nowhere near zero: its minimum over those sectors is 10 below $M = 3000$ and 520 for M near 2×10^4, against 497 in the ordinary classes.

That calculation has been carried out, and on our reading it closes the local route. Coupling consecutive sectors and admitting the lines $11, 13, 17$ bounds the exception budget over a full 210-window by 31 under the twinless hypothesis. No finite set of lines can lower that ceiling. Of the 28 maximal configurations, 27 are killed by a fixed prime divisor once the quadratic partner conditions are included. The survivor is admissible at every prime, so the residues can be chosen simultaneously against any finite list: the setting of Schinzel's Hypothesis H [19]. The order in which the lines are born adds nothing either. The calculation and its verifications are in [P6, §2.3]; here we record only what it means.

So the upper-bound side is finished, and the whole difficulty sits in the other half. One need not kill the escapee: if $\sum_i C_{M_i} \geq 32$ can be established, the thirty-second survivor has no room inside E and is a twin. Measured, that requirement is met with enormous room: $B_k := \sum_i C_{M_i}$ is 1,337 at $k = 1$, 29,579 at $k = 100$ and 196,948 at $k = 1000$, with $B_k \log^2 M_0/M_0$ flat at about 139 from $k = 20$ onward, and the minimum over a band of sixty consecutive k never below 1,337. **The requirement is exceeded by a factor of six thousand and rising, and remains unproved.**

And lengthening the window does not help. Sieving an interval of length H to depth z has $s = \log H/\log z$. For one sector $H \asymp 12M_0$ and for the full window $H \asymp 420M_0$, both against $z \asymp M_0$. So $s \to 1$ in each, and the gain is $\Delta s = \log 35/\log M_0$: 0.257 at $M_0 = 10^6$, 0.129 at 10^{12}, vanishing. Against the two-dimensional sifting limit $\beta_2 = 4.2664$ this closes under eight per cent of the gap (0.257/3.27), and less as M_0 grows: **a constant multiple of the length buys nothing asymptotically, only a power of M_0 would.** The reorganisation is a clarification, not a reduction, but a real one: it closes one of the two branches permanently and by proof.

So the requirement is now one survivor, in one arithmetic progression, for infinitely many M — and it is still the same wall. To be bounded below: the number of cells of an interval of length $\asymp 2M$ surviving every line $p \leq M$, a two-dimensional sieve at $u = 2$, where the sifting limit is $\beta_2 = 4.2664$ and no positive lower bound is available. Lowering the demand from $C_M \geq 7$ to $C_M \geq 3$ to $C_M \geq 1$ does not approach that bound; the sieve supplies none of the three. **The reduction is worth recording because it is so sharp and still does not bind: it separates what the framework can do — fix the arithmetic of the exceptional positions completely — from what it cannot, which is to produce a single guaranteed survivor.**

A remark from the same literature, which corroborates §2. Campbell records that reaching $\Omega \leq 2$ between squares appears to lie beyond his framework because it uses only Type I information, however large n is taken [5]; Chen's asymptotic $k = 2$ [6] comes from a different route. What remains on this side is the numerical optimisation itself, and at $k = 2$ even that is not enough. The *bilinearity* deficit of [P11, §2.1] is thus the one the specialists name too: the fourth of the four, met in a test problem where the first three do not bite.

2.8 A third test case: a shared cofactor on two lines

Look at the cofactors, not the numbers. Two lines L_p and L_q **share** the cofactor a when pa and qa both lie in one window between consecutive odd squares. One natural condition on a is needed: both strikes come after both lines have begun.

> **Theorem 1.** Let $p < q$ be odd primes and let $a \geq q$ satisfy $k^2 < pa < qa < (k+2)^2$ for some odd k. Then $q = p + 2$. (The hypothesis $a \geq q$ is not a restriction: it holds automatically whenever $q \leq k$, and is what confines the sharing to the lines above $k/2$ — see below.)

Proof. Both $\sqrt{pa}$ and $\sqrt{qa}$ lie in $(k, k+2)$, so their difference is less than 2. That difference is

$$\sqrt{qa} - \sqrt{pa} \;=\; \frac{(q-p)\sqrt{a}}{\sqrt{q}+\sqrt{p}} \;>\; \frac{(q-p)\sqrt{q}}{2\sqrt{q}} \;=\; \frac{q-p}{2},$$

using $a \geq q$ for the numerator and $p < q$ for the denominator. Hence $q - p < 4$, and the difference of two odd primes is even, so $q - p = 2$. ■

Nothing asks a to be prime, or the strikes to be **primitive** (with p the least prime factor of pa).

An inherited strike is an equally good witness: at $k = 31$,

$$31^2 \;<\; 29 \cdot 35 = 1015 \;<\; 31 \cdot 35 = 1085 \;<\; 33^2,$$

The cofactor 35 is composite and both strikes are owned by smaller lines, yet the pair $(29, 31)$ is still exhibited. Primitive versus inherited matters for ownership and coverage counts, as in [P9, §2.2], not here.

Verification. Over all windows with odd $k < 1000$: 5,593 sharing pairs with $a \geq q$, of which 4,764 have a composite cofactor, and **not one** has $q - p \neq 2$.

The converse is immediate, with an explicit cofactor. For any twin $(p, p+2)$ take $a = p + 6$. Then

$$p(p+6) = (p+2)^2 + 2p - 4, \qquad (p+2)(p+6) = (p+4)^2 - 4,$$

so both strikes lie in $((p+2)^2, (p+4)^2)$ for every $p > 2$, whatever the factorisation of $p + 6$. No third prime and no congruence condition is needed: every twin shares a cofactor in the window whose root is its upper member plus two.

> **Theorem 2.** Infinitely many twins exist if and only if two odd primes share a cofactor, in the sense of Theorem 1, in windows of unbounded height.

The horizon. A fixed twin has finitely many sharing windows. The constraints $k^2 < pa$ and $(p+2)a < (k+2)^2$ require $k^2/p < (k+2)^2/(p+2)$, that is $(p+2)/p < (1+2/k)^2$, which gives $k < 2p$ asymptotically. Measured: the largest window root is $1.824p$ at $p = 17$, $1.901p$ at $p = 71$ and $1.944p$ at $p = 269$, approaching the bound from below.

2.9 The geometry of sharing: where it sits, and how it is forced

In the window $W_k = (k^2, (k+2)^2)$ the line L_p has the cofactor set $C_p = \{a \text{ odd} : k^2 < pa < (k+2)^2\}$, an interval of length $(4k+4)/p$. Two such sets meet only if $q/p < (1+2/k)^2$, which for $q = p+2$ is $p > k^2/(2k+2)$: **all sharing lives in the upper half** $k/2 < p < q \leq k$; no line below $k/2$ takes part. There the hypothesis $a \geq q$ of Theorem 1 is automatic, since $a > k^2/q > k \geq q$, and may be dropped. The overlap of C_p and C_{p+2} has length

$$\frac{(4k+4)p - 2k^2}{p(p+2)},$$

a function of p/k alone: 0.17 at $p = 0.51k$, 1.12 at $0.6k$, 1.88 at $0.8k$ and exactly 2 at $p = k$. Since the sets are intervals and the overlap never exceeds 2, **two lines share at most one odd cofactor, and it is the first cofactor of the smaller line and the last of the larger.** In $(19^2, 21^2)$: line 11 has $33, 35, 37, 39$, line 13 has $29, 31, 33$, line 17 has $23, 25$, line 19 has $21, 23$; the two sharings are 33 and 23, each at the ends. Over all 2,604,125 pairs of odd lines in windows with odd root below 501 there are 12,019 sharings. Each is a single cofactor, the first of the smaller line and the last of the larger, with $k/2 < p < q \leq k$.

Sharing can also be produced outright. Write $k = p + 2t$ with $t \geq 1$ and take

$$a = p + 4t + 2. \qquad \text{Then} \qquad pa = k^2 + 2p - 4t^2, \qquad (p+2)a = (k+2)^2 - 4t^2,$$

so both strikes lie in W_k **whenever** $2t^2 < p$, i.e. whenever the two lines are close enough to the root. Verified on 44,551 instances of the identity and 3,168 of the inclusion, without exception. So near

the root, sharing between adjacent odd lines is a construction, not a tendency; measured, 92.5% of all adjacent odd pairs above $k/2$ share, at $k = 10^3, 10^4, 10^5$ alike.

The cofactor lists move by a law, exact inside the candidate strip. The list $F_p(k) = \{a \text{ odd} : k^2 < pa < (k+2)^2\}$ is the strike set of L_p read on the cofactor axis. On the cell axis [P2, Cor 4] presents it as two interleaved progressions of step p; on this axis it is a single run. For a candidate line — one in the upper half $k/2 < p \le k$ where all sharing lives — the run has $h_p(k)$ terms with $2 \le h_p \le 4$. This is forced: k and p are odd, so $p \ge (k+1)/2$ and the interval has length $(4k+4)/p \in (4, 8]$, which holds between two and four odd integers, consecutive by definition. Measured over 40,191 lines, with $h_p = 2, 3, 4$ occurring 15,414, 18,707 and 6,070 times. With A_p and B_p its first and last entries, $p < q$ gives $A_p > A_q$ and $B_p > B_q$: the runs slide left as the line grows. The overlap never exceeds one entry, so sharing is an equality of endpoints,

$$F_p(k) \cap F_q(k) \neq \varnothing \iff A_p = B_q \iff A_p - A_q = 2(h_q - 1),$$

verified on all 2,666,595 pairs of candidate lines with $k < 801$.

In the next window the line resumes where it stopped, except that a strike exactly on the shared boundary belongs to neither open window and is skipped:

$$A_p(k+2) = B_p(k) + 2 + 2 \cdot \mathbf{1}_{\,p \mid (k+2)^2} = A_p(k) + 2h_p(k) + 2 \cdot \mathbf{1}_{\,p \mid (k+2)^2},$$

exact on all 79,790 line-window pairs tested, of which 1,398 trigger the indicator. **And the indicator never fires on a prime line of the strip.** If p is prime and $p \mid (k+2)^2$ then $p \mid k+2$. As $k+2 > p$ and both are odd, $k+2 \ge 3p$, while $p > k/2$ gives $3p > 3k/2 > k+2$ for $k \ge 5$, a contradiction. Checked: none of the 77,427 prime lines in the strip with $k < 3001$ triggers it, and all 128 occurrences inside the strip below $k = 801$ are on composite lines.

So for prime candidate lines the movement is an exact equality. Each line advances by $2h_p$, which is 4, 6 or 8, and for two lines the difference of the run-starts obeys

$$D_{p,q}(k+2) = D_{p,q}(k) + 2\big(h_p(k) - h_q(k)\big),$$

changing by 0 or ± 2 or ± 4 at each step, with sharing exactly when $D_{p,q}$ reaches $2(h_q - 1)$. This is the sharpest description of the motion the framework produces. The boundary indicator is the first quantity in this work that tells prime lines from composite ones outright: it fires only on composite lines. But it separates them in the wrong direction. It says which lines can break the movement law, not which can share. The sharing condition remains a statement about the runs, and the runs do not know whether their index is prime.

The sharing condition in closed form, and why it is blind. The quadratic walk turns the criterion into one inequality. In the window $k = p + 2s$ with $s \ge 1$, the sharing of the lines p and $p+2$ is

$$\left\lfloor \frac{2s^2}{p} \right\rfloor = \left\lfloor \frac{2s^2}{p+2} \right\rfloor \iff (2s^2 \bmod p) \ \ge \ \frac{4s^2}{p+2},$$

the second form from writing $2s^2 = jp + r$: the two floors differ exactly when $r < 2j$. **The index is s, not $s+1$**, and the window $s = 0$ has no sharing: there $F_p = \{p+2, p+4\}$ while the cofactors of $p+2$ all lie below it. At $s = 1$ the residue is 2 and the threshold below 1, so sharing occurs in the first window where it can, $\big((p+2)^2, (p+4)^2\big)$, with cofactor $p+6$: the witness above, recovered independently.

One boundary case separates the floor test from sharing itself, and it is where primality enters. The floor equality is blind to the upper endpoint of the open window: it counts a candidate cofactor a with $(p+2)a = (k+2)^2$, which the window excludes. That happens exactly when $p+2$ divides $(k+2)^2$. Since $k+2 \equiv 2s \pmod{p+2}$, this is $(p+2) \mid 4s^2$, impossible for $p+2$ prime with $s \le (p-1)/2$, and the only way the two can differ. Hence

$$\text{floors equal} \iff \text{sharing}, \qquad \text{for every } p \text{ with } p+2 \text{ prime},$$

while for composite $p+2$ the floor set can be strictly larger. Over all odd $p < 400$ and every s there are 19,899 pairs and exactly 93 such cases, all at composite $p+2$, all a floor equality with no sharing. Regenerated by `code/verify_sharing_index.py`. *(So the set S_p of §2.11 is defined by the floors, and equals the set of sharing windows exactly in the case the criterion is applied to.)*

This closed form explains the measured rate. If $2s^2 \bmod p$ is equidistributed, sharing at s has probability $1 - 4s^2/p^2$, and averaging over $s \le p/2$ gives $1 - \frac{1}{3} = \frac{2}{3}$. Measured, the rate is 0.6654 over pairs of prime lines and 0.6662 over all other adjacent odd pairs. **It also explains why the criterion cannot see primality: it depends on the distribution of $2s^2 \bmod p$, not its structure, and the distribution is the same for prime and composite p.**

The structure does differ. The map $s \mapsto s^2 \bmod p$ is injective on $[1, (p-1)/2]$ **if and only if p is prime** (a collision $s_1^2 \equiv s_2^2$ needs $p \mid (s_2 - s_1)(s_2 + s_1)$ with both factors below p), with the single exception $p = 9$, where the smallest collision falls outside the range. Checked: injective for all 76 odd primes below 400 and for exactly one of the 122 odd composites. But the deviation of the sharing count from its mean does not detect it. Measured as $|\cdot|/\sqrt{p}$ over $p \in [8000, 20000)$, it is 0.270 for primes, 0.260 for composites with least factor at least 11, 0.222 for multiples of 3 and 0.203 for multiples of 5: the same order throughout, and *smaller*, not larger, for composites with small factors. The sharing count is a sum over single values of s with a threshold that moves with s, and the sum averages away any local structure in the residues.

And that is exactly where primality is missing. Take $k = 23$. The cofactor lists are $13 : 41, 43, 45, 47$; $17 : 33, 35$; $19 : 29, 31$; $21 : 27, 29$; $23 : 25, 27$. The construction forces two sharings, lines $19, 21$ at the cofactor 29 and lines $21, 23$ at the cofactor 27, and all four strikes lie in $(529, 625)$. But the shared line is 21, which is composite. The two prime lines, 19 and 23, use *different* cofactors in these sharings, and the composite line between them does not merge them. Of the 25,001 adjacent odd pairs above $k/2$ at $k = 10^5$, 23,069 share and only 519 consist of two primes — a ratio of 2.1%, falling like $1/\log^2 k$.

2.10 Why the sharing equivalence closes the route

The proof of Theorem 1 used the primality of p and q at one point only, to say that $q - p$ is even. Everything geometric in it holds for any two odd numbers two apart:

$$23^2 \;<\; 21 \cdot 27 = 567 \;<\; 23 \cdot 27 = 621 \;<\; 25^2,$$

and 21 is not prime. So the square law forces a shared cofactor between **adjacent odd numbers**. What is missing is precisely that infinitely many adjacent odd pairs are both prime: the conjecture itself. To prove "sharing occurs at unbounded height" one must exhibit the two primes, since sharing between primes is defined by their primality and cannot be stated without it.

The contrast with the sector criterion [P6, §2.2] is one of degree, not kind. Primality is itself a survival statement ($N < x$ is prime exactly when it survives every line below $\sqrt{N}$), so no reformulation

moves the problem out of the sieve's language, and none in this work ever did. What changes between criteria is one number: the ratio s of the logarithm of the range to the logarithm of the sifting depth at which a survivor is asked for. A dimension-two lower bound needs $s > \beta_2 = 4.2664$. The sector criterion [P6, §2.2] asks at $s \approx 1$, and the sharing criterion at $s \approx 2$, since it needs a survivor to depth $\sqrt{N}$ in a range of length N. Below the threshold the sieve returns zero, not a small bound, so neither is available. **What the sieve cannot do is not recognise a prime — sifting to $\sqrt{N}$ recognises it exactly — but prove that a survivor exists at these depths.** An equivalence between criteria helps only when it raises s. This one raises it from 1 to 2, the largest gain in this work, and still falls short of the threshold by more than it gained.

This is the sharpest form the obstruction takes in this work. A criterion built from a one-parameter sum over n sees only the distribution of $2n^2 \bmod p$, and that distribution is prime-blind. What distinguishes a prime modulus is the absence of *collisions*, a statement about **pairs** (n_1, n_2). Every criterion in this paper is a one-parameter sum, which is why each turned out equivalent to the conjecture rather than weaker. The need for a bilinear decisive input — a statement about pairs — is the **Bilinearity** deficit of [P11, §2.1], reached here from the framework's local coordinates rather than from the sieve literature.

2.11 A fourth test case: collisions inside the sharing set

The sharing criterion of §2.8 can be pushed one step further. The step yields a criterion that *reads* the primality of $p+2$ instead of assuming it, and then fails for a reason sharper than any above.

Fix an odd p and write $N = p+2$. Number the windows by s, with root $m = p + 2s$. The sharing condition of §2.9 becomes a comparison of two floors, and the set is

$$S_p \;=\; \Big\{1 \le s \le \tfrac{p-1}{2} \;:\; \Big\lfloor \tfrac{2s^2}{p} \Big\rfloor = \Big\lfloor \tfrac{2s^2}{N} \Big\rfloor \Big\} \;=\; \Big\{ s : (2s^2 \bmod p) \ge 2\lfloor \tfrac{2s^2}{p} \rfloor \Big\},$$

the set of sharing windows exactly when N is prime, by the boundary case of §2.9. Its size carries no information: the density is near $2/3$ whether N is prime or not. What carries information is a **collision**: a pair $u < v$ in S_p with $u^2 \equiv v^2 \pmod N$. Write E_p for the number of such pairs. *(Here u, v are two members of S_p, unrelated to the composite rates u, v of [P9, §2.3].)*

Theorem 3. For odd $p > 7$: $E_p = 0$ if and only if $p+2$ is prime.

One direction is immediate: if N is prime it divides neither $v-u$ nor $v+u$, both positive and below N. The work is in the other direction. That an odd composite $N = ab$ admits *some* pair $u = (b-a)/2$, $v = (b+a)/2$ with $v^2 - u^2 = N$ is the classical difference-of-squares criterion, in use since Fermat; **the content here is that the witness can always be taken inside S_p.** The argument has five steps: (i) if $v^2 - u^2 = \mu N$ and $v \in S_p$ then $u \in S_p$, by subtracting $2\mu N$ from the double inequality defining membership, so only the larger member must be placed; (ii) with $k = \lfloor b/2a \rfloor$, $c = 2a(k+1) - b$ and $\Delta = bc - a^2$ one has the identity $2v^2 = (k+2)N - \Delta/2$, and the witness fails exactly when $0 < \Delta \le 4k+8$; (iii) that forces $b > 3a$, since $a < b \le 3a$ gives $3N - 2v^2 = (3a^2 - (b-2a)^2)/2 \ge a^2 > 6$ and both floors equal 2; and since Δ increases with c on $0 < c < 2a$, failure also forces $2a(k+1) - 1 - a^2 \le 4k+8$, that is $k(2a-4) \le a^2 - 2a + 9$; (iv) the split $(3a, b)$ of $3N$ then works. Here $2v_3^2 = (k+4)N + R$ with $R = 4a^2 - \Delta/2$. For $k \ge 2$ one has $N - 2(k+5) - R = (2k - \frac{5}{2})a^2 + c(ak - \frac{c}{2}) - 2k - 10 > 0$ and, by the bound on k, $R \ge 4a^2 - 2k - 4 > 0$, so both floors equal $k+4$. For $k = 1$ that bound fails, but then $R - N = (a-c)^2/2$, so $2v_3^2 = 6N + (a-c)^2/2$ with $0 < (a-c)^2/2 < a^2/2 < N - 14$, and both floors equal 6; this branch is the Pell family $(b-2a)^2 - 3a^2 = -2$, that is $(a, b) = (3, 11), (11, 41), (41, 153), \ldots$, first met at

$p = 31$; (v) for $N = q^2$ the pair $(q, 2q)$ serves, with $v^2 - u^2 = 3N$. Each step is proved above. As a check, `code/verify_proved_identities.py` confirms the identity of (ii), the failure criterion and the landing of the rescue on every factorisation it reaches (to $p = 3 \cdot 10^6$ under `--full`: 112 failing factorisations at 111 odd values of p, 32 of them at prime p, five on the $k = 1$ branch, and every rescue inside S_p).

2.12 Where the collision must sit, and what a narrow slice sees

A witness can always be taken inside a belt.

> **Theorem 4.** Let $p > 121$ be odd with $p + 2$ composite. Then some collision in S_p has $\sqrt{p+2} < v \le (p + 11)/6$.

The upper bound is $6v \le N + 9 \iff (a - 3)(b - 3) \ge 0$, which holds for every factorisation with $a \ge 3$. The rescue witness stays inside because $6v_3 \le N + 9 \iff (a - 3)b \ge 9(a - 1)$. This follows from $b > 3a$ for $a \ge 7$, from $b \ge 25$ for $a = 5$, and for $a = 3$ from the fact that the ordinary witness then fails only at $p = 31, 49, 67, 85, 103, 121$ — a finite list, because step (iii) of Theorem 3 gives $k \le 6$ when $a = 3$, so $b \le 41$. The lower bound is forced by $v^2 - u^2 \ge N$. **The hypothesis is that p be odd, not prime**: among odd $p > 7$ the only exceptions are $23, 31, 47, 49, 67, 85, 119, 121$, of which the primes are $23, 31, 47, 67$. Reading the bound through $v - u = a$ and $v + u = b$ shows what the belt is. A balanced factorisation puts the collision near $\sqrt{N}$, a lopsided one near $N/6$; the belt is the image of all factorisations of N on the axis of window positions. Only v is confined: at $p = 999{,}917 = 991 \cdot 1009 - 2$ the witness is $u = 9$, $v = 1000$.

And a slice at the top of the belt collapses to a divisibility test.

> **Theorem 5.** Let $p > 121$ and $H = (N + 9)/6$. Inside the slice $H - L \le u < v \le H$ with $3 \le L < \sqrt{N/3}$, a collision exists if and only if $3 \mid N$, and then it is the single pair $u = (N - 9)/6$, $v = (N + 9)/6$.

This follows from the identity

$$N\ (v - u - 3k)\ =\ 3\Big[\big(\tfrac{N}{6} - u\big)^2 - \big(\tfrac{N}{6} - v\big)^2\Big],$$

whose right side has modulus below N inside the slice while the left is a multiple of N. Hence $v - u = 3k$, then $u + v = N/3$, so $3 \mid N$, and $v \le N/6 + 3/2$ forces $k = 1$. Verified over the primes in $(121, 6 \cdot 10^4)$: no collision when $3 \nmid N$, and exactly one in each of the 3,001 cases with $3 \mid N$.

Collisions enter the slice in numbered stages, each with an explicit threshold. Keep $H = (N + 9)/6$ and the slice $H - L \le u < v \le H$. For a collision write $d = v - u$ and $s = u + v$, so $v^2 - u^2 = kN$ becomes $ds = kN$. As N is odd, d and k have the same parity, so $d - 3k$ is even; put $d - 3k = 2t$. Then

$$2tN\ =\ d\ (N - 3s),$$

and the lower edge of the slice gives $s \ge 2(H - L) + d$, whence $2tN \le 3d(2L - 3 - d)$. The right-hand side is $3[(L - \frac{3}{2})^2 - (d - L + \frac{3}{2})^2]$ by an identity, so the bound holds for every d with no further hypothesis:

> **Theorem 6.** For $p > 121$ and $L \ge 3$, every collision inside the slice satisfies $0 \le t \le \Big\lfloor \dfrac{3\ (L - \frac{3}{2})^2}{2N} \Big\rfloor$, $\qquad t = \frac{1}{2}(d - 3k)$. More generally, for a slice with top H_0 the bound reads $t \le \lfloor (N - 6H_0 + 6L)^2/(24N) \rfloor$.

The stages open one at a time as the slice widens: $t = 0$ needs only $L \geq 3$ and is the pair of Theorem 5; $t = 1$ needs $L \geq \frac{3}{2} + \sqrt{2N/3}$; $t = 2$ needs $L \geq \frac{3}{2} + \sqrt{4N/3}$. **This widens Theorem 5 by a factor $\sqrt{2}$:**

$$3 \leq L \ < \ \tfrac{3}{2} + \sqrt{2N/3} \quad \Longrightarrow \quad \text{the slice detects divisibility by 3 and nothing else.}$$

The parity law and the identity were checked on 276,412 collisions without exception. The bound held on every slice tested once the slice top and the reference point use the same H: the exact rational $(p + 11)/6$, not its integer part. This matters at small p and produced spurious violations before it was fixed. The census bears out the staging: in the last 2% of the belt below $p = 1.2 \cdot 10^5$ there are 5,560 collisions at $t = 0$, 1,027 at $t = 1$, and none above.

The earlier counterexample is now explained, not just recorded. At $N = 64{,}375$ the pair $u = 10{,}522$, $v = 10{,}728$ has $d = 206$, $k = 68$, hence $t = 1$; the narrowest slice containing it has $L = H - u = 208.6667$, while the stage-one threshold is $\frac{3}{2} + \sqrt{2N/3} = 208.6634$. It clears the threshold by 0.0033.

What the staging does not give. A slice below the stage-one threshold sees only divisibility by 3. A vanishing count there says nothing about the primality of $p + 2$, since the witness for a different factor lives in the part cut away. The staging law explains the mechanism of every relaxation tried above (which factors can reach which depth) without turning any of them into a criterion.

2.13 What a first moment on the collision count would need

E_p is integer-valued and vanishes exactly on twins, so $E_p \geq \mathbf{1}_{\{p+2 \text{ composite}\}}$ pointwise, and for *any* non-negative weights w,

$$\sum_p w(p) E_p \ \geq \ \sum_p w(p) - \sum_{p \text{ twin}} w(p).$$

So $\sum wE < \sum w$ **implies** $\sum_{\text{twin}} w > 0$. **That makes it a sufficient condition, and a legitimate way to force a zero**: for an integer-valued non-negative statistic, an upper bound on the mean below one produces a vanishing value, and arguments of exactly that shape are standard.

An earlier version of this subsection read the implication as circularity and concluded that every weighted first-moment argument assumes its conclusion. That was a mistake in logic, corrected here. Proving a sufficient condition is not assuming it. What is true, and is the whole difficulty, is narrower: **no upper bound on $\sum wE$ is available from inside this framework**. E_p is defined from the factorisation of $p + 2$, so any bound on its mean is already a statement about how often $p + 2$ factors; the sieve delivers rough numbers, not that. The same applies to any statistic vanishing exactly on twins and bounded away from zero elsewhere, which Theorem 4 guarantees for $1/v$. Higher moments do not supply it either: Paley–Zygmund bounds $\mathbf{P}(E > 0)$ from *below*, and the measured value of that bound is near 0.46 throughout.

So this route is open and unequipped, not closed. What would close it is an upper bound for $\sum_p w(p) E_p$ obtained independently of the twin count. The four relaxations below say why the obvious way to get one, widening the zero set, does not work.

Relaxing the statistic so its zero set is *wider* than the twins is the natural repair. It was tried in four independent families: thresholding the recovered factor, restricting the distance $v - u$, cutting the top of the belt, cutting the bottom. In each, the zero set widens faster than the mean falls: at a mean near 1 the extra family is already several times the number of twins. Two honest limits on that reading. The mean falling below 1 is *sufficient* to force a zero, not necessary for every analytic argument. And it does not by itself produce the collapse of Theorem 5, whose proved width $\sqrt{N/3}$

is narrower than the width $\sqrt{2p/3}$ that a mean below 1 permits. The gap between the two is real and inhabited: at $p = 64{,}373$, with $N = 625 \cdot 103$ not divisible by 3, the pair $u = 10{,}522$, $v = 10{,}728$ lies in S_p and in the slice, colliding with $k = 68$ and exposing the factor 103.

What is established is the complete local geography: a criterion equivalent to the primality of $p + 2$, an exact belt for its witness, and an exact collapse in a narrow slice. What is absent is any relation between the belt of one prime and the next. The factorisations of $p + 2$ and of the next $q + 2$ are independent, and the measured record bears this out. Among the primes below 10^6 the smallest factor of $p + 2$ reaches 991, no fixed set of small lines accounts for all collisions, and a run of 102 consecutive primes from 850,351 all have a collision.

2.14 Counting the witnesses, and why a count of them cannot close the gap

The collisions of §2.11 can be counted exactly, not estimated, and the count turns out to re-encode the collisions themselves. Fix a prime $p > 121$ with $3 \nmid N$, write $X = N - 6u$ as before, and set

$$\Delta \;=\; (N - 6v)^2 + 48t, \qquad \text{so that} \qquad X^2 - \Delta \;=\; 24tp.$$

> **Theorem 7.** Within one list S_p, among the collisions of the belt — those with $v \le H = (N + 9)/6$, which by Theorem 4 always include a witness — the value of Δ determines the collision: two such collisions with the same Δ are the same pair.

Proof. Subtracting the two identities gives $(X_1 - X_2)(X_1 + X_2) = 24p(t_1 - t_2)$. The belt restriction puts $X_i = N - 6u_i$ strictly between 0 and p: $u_i < v_i \le (N + 9)/6$ gives $X_i > N - (N + 9) + 6 > 0$, and $u_i \ge 1$ gives $X_i \le N - 6 = p - 4$. Both X_i are odd and in $(0, p)$, so the difference is below p and the sum is an even number below $2p$. Neither is divisible by the prime p unless the difference vanishes. Hence $X_1 = X_2$, so $u_1 = u_2$ and $t_1 = t_2$; equal Δ then forces $(N - 6v_1)^2 = (N - 6v_2)^2$, so either $v_1 = v_2$ or $v_1 + v_2 = N/3$, and the second is impossible when $3 \nmid N$. ■

Verified over the 267,575 collisions of the belt on 1,121 prime lines below $2 \cdot 10^4$: no value of Δ repeats inside any list. **So within the belt the number of collisions equals the number of distinct values of Δ; recoding each collision by a single number relaxes nothing.**

The belt restriction is load-bearing in the proof and not known to be removable. Off the belt X leaves the range the proof needs: at $p = 131$ the collisions $(25, 32)$ and $(44, 51)$ of S_p give $X = -17$ and $X = -131$, the second divisible by p itself. A search to $p = 2 \cdot 10^4$ found no repeated Δ off the belt either, so the wider statement may well be true; only the narrower one is established here. The prohibition of §2.13 removes shapes that would contradict the primality of p; it removes no valid collision at a prime line.

A constructive counting bound follows. Each Δ satisfies $\Delta \equiv 1 \pmod{24}$ and is never a square, so the admissible values begin $73, 97, 145, 193, 217, 241$, with the squares $1, 25, 49, 121, 169$ excluded. In a slice whose first admissible position is $U = \max(1, \lceil H - L \rceil)$, every collision has $u \ge U$ and $t \ge 1$, whence

$$\Delta \;\le\; M \;=\; (N - 6U)^2 - 24p,$$

and the number of collisions is at most the number of non-square integers congruent to 1 modulo 24 below M. If $M < 73$ the slice is empty; at $M = 73$ there is one candidate, at $M = 97$ two. Direct enumeration on 786 slices found no case above the bound. Four instances:

p	slice width L	admissible Δ	collisions
131	10	0	0
389	18	1	1
64,373	209	2	1
64,373	210	587	1

And the first row is the limit of the whole approach. At $p = 131$ the slice is provably empty, yet $N = 133 = 7 \cdot 19$ is composite: the witness is $u = 6$, $v = 13$ with $13^2 - 6^2 = 133$, both in S_p, and the slice begins above them, at $U = 14$. The emptiness is an artefact of where we looked.

So the requirement on any relaxation is now exact: **it must retain at least one witness for every composite $p + 2$.** The counting bound does not guarantee that, so it cannot carry an empty slice to a twin. §2.13 reaches the same conclusion from means and weights; here it is reached by construction: a bound that proves zero, and an example showing the zero can be false.

2.15 The pattern of the whole work

Across the tested routes, the local laws are much sharper than the available summed bounds.

The routes examined here show this repeatedly: the pigeonhole constraint ([P11, App. B.2.2]), the deletion budget ([P11, App. B.2.1]), the criterion $K < G$ ([P11, App. B.2.1.3]), the collective cancellation [P4, §4], and the covering condition for Jacobsthal (§2.4). Each time a sharp local law is followed by a summation (equivalently, a passage from cycle to short window) whose loss exceeds the available margin. **[P3, §5.2] isolates why with unusual clarity: making the layers disjoint removes the double-counting entirely, so the summed bound stops diverging and stays below 1 for every depth — yet it then returns exactly $\prod(1 - 1/q)$. The interference was never in the intersections; it lives in the size of each layer, which already contains the product of all the layers before it. We therefore treat this as the recurring limitation revealed by the present tests, not as a theorem excluding every possible summation method.**

The computations and much of the prose in this paper were prepared with AI assistance (ChatGPT, OpenAI; Claude, Anthropic), used for algebraic derivation, for drafting and rewriting code and text, for running the computations, and for auditing the papers against their own scripts. All statements were checked by the author, who is responsible for them; the repository README sets out the division of labour in full.

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